# A Classification-Based, Semianalytical Approach for Estimating Water Clarity From a Hyperspectral Sensor Onboard the ZY1-02D Satellite

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***Abstract*— Water clarity ($Z_{sd}$) is a widely used quality indicator that can be estimated from remote sensing imagery. China's newest generation advanced hyperspectral imager (AHSI) onboard the ZY1-02D satellite is expected to enable accurate water clarity retrieval for inland waters, since AHSI can provide abundant band choices, while its 30-m spatial resolution is advantageous for monitoring small inland water bodies. In this study, to retrieve $Z_{sd}$ from the ZY1-02D imagery for inland waters with varying turbidities, we propose a classification-based, semianalytical method, in which the red/blue band ratio is employed to distinguish clear to moderately turbid water and highly turbid waters. Two quasi-analytical algorithms (QAAs), $QAA_{v5}$ and $QAA_{m14}$, are used to estimate the total absorption coefficient [$a(\lambda)$] and the backscattering coefficient [$b_b(\lambda)$] for clear to moderately turbid water and highly turbid waters, respectively. The estimated $a(\lambda)$ and $b_b(\lambda)$ are utilized to obtain the diffuse attenuation coefficient $K_d$, followed by the $Z_{sd}$ calculations. Compared with 70 matchups of *in situ* measured $Z_{sd}$ values (0–6.5 m), the ZY1-02D image-derived $Z_{sd}$ achieved an $R^2$ of 0.98, with an average unbiased relative error and root mean square error of 29.1% and 0.52 m, respectively. In addition, the proposed method can yield $Z_{sd}$ with higher accuracies than that of optimized empirical models. Therefore, the ZY1-02D AHSI imagery can retrieve reliable $Z_{sd}$ for both clear (>3 m) and turbid waters (0–3.0 m), thereby serving as a useful satellite data source for monitoring the water clarity of large-scale inland water bodies.**

***Index Terms*— Hyperspectral remote sensing, quasi-analytical algorithm (QAA), Secchi disk depth, semianalytical method (SAM).**

## I. Introduction

WATER clarity is an important quality indicator that is generally quantified using the Secchi disk depth ($Z_{sd}$). $Z_{sd}$ is measured by lowering a Secchi disk into water until it can no longer be seen. Water clarity depicts how deep the sunlight penetrates the water, which affects the growth of aquatic plants and algae [1], [2]. It also helps in evaluating the trophic state of water bodies owing to its link to water constituents, such as Chlorophyll-a. Traditionally, water clarity is measured during ship surveys or at ground stations. Despite the easy operation of field measurements, conducting large-scale and repetitive measurements is costly and time-consuming. In contrast, optical remote sensing data provide an avenue for efficiently acquiring large-scale and long-term water clarity data.

Generally, $Z_{sd}$ is retrieved from remote sensing data based on empirical models or semianalytical methods (SAMs). Empirical models can be established from regression analysis of single band, band ratios, or band combinations [3]–[6]. This method is easy to perform and can usually provide reliable results for a certain waterbody but tends to fail when applied to a new region with different optical properties. Conversely, SAMs are based on the estimation of the absorption and backscattering coefficients [7]–[9], which are typically calculated using quasi-analytical algorithms (QAAs). Their effectiveness has been shown in oceanic, coastal, and inland waters [10], [11]. Consequently, various versions of QAAs have been developed [12]–[14] to produce the best estimates for the absorption and backscattering coefficients of waters with diverse optical properties. To retrieve accurate $Z_{sd}$ for different water types, a feasible technique is to first classify waters and then apply the most suitable QAA to those classes [15], [16].

In 2018, China launched the GF-5 satellite with a 330-band hyperspectral instrument, known as the advanced hyperspectral imager (AHSI) [17]. The spectral resolution of the AHSI is ~5 nm for the visible-to-near-infrared (VNIR) bands and 10 nm for the short-wave infrared (SWIR) bands; it can acquire 30-m spatial resolution imagery with a 60-km swath.

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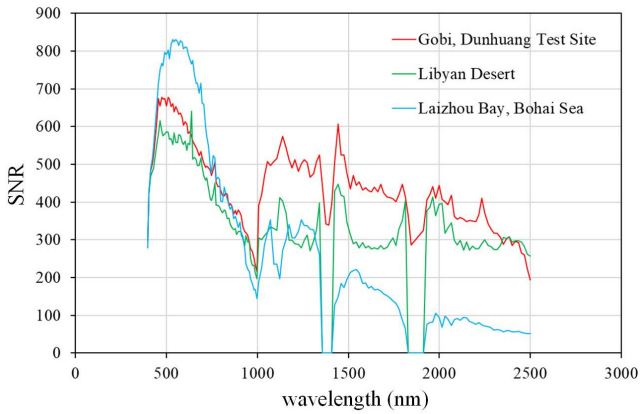



Fig. 1. SNR of various surface types in ZY1-02D AHSI images: Gobi at Dunhuang Test Site (March 4, 2020), Libyan Desert (December 23, 2019), and Laizhou Bay, Bohai Sea (January 9, 2020).

The AHSI imagery of GF-5 has been successfully applied for the water clarity monitoring of inland waters [18]. The ZY1-02D satellite was subsequently launched by China on September 12, 2019, operating at an altitude of 778 km with two payloads (i.e., the newest generation AHSI and a nine-band multispectral imager). The revisit cycle of ZY1-02D is 55 days, but the revisit frequency for a place of interest can be shortened to three to five days in the off-nadir mode.

The newest AHSI onboard ZY1-02D satellite has the same spatial resolution and swath as that on the GF-5 satellite. Its 60-km swath is wider than previous hyperspectral sensors (e.g., Hyperion) and provides better coverage for inland water bodies. In addition, for the AHSI of ZY1-02D, the number of bands is reduced to 166 to improve the signal-to-noise ratio (SNR), thereby decreasing the spectral resolution to 10 nm for the VNIR bands and 20 nm for the SWIR bands. According to the on-orbit performance evaluation of the ZY1-02D satellite, the SNR on various surface types was calculated as the ratio of the mean value to the standard deviation of radiance (or gray-level count) from a spatially homogenous target [19], [20]. SNRs were compared among homogenous gobi, desert, and oceanic water (see Fig. 1). The SNR of the oceanic scene is higher than the desert and gobi surfaces in the spectral range from 400 to 1000 nm but can be higher than 700 for the ZY1-02D bands from 450 to 700 nm.

The AHSI imagery from the ZY1-02D satellite is seen to be a useful data source for high-precision $Z_{sd}$ estimation of inland waters. First, compared with the widely used Landsat series, AHSI provides more band choices for water clarity retrieval. Moreover, the bandwidth of AHSI is narrower, such that the absorption and scattering of different substances in the water are not aggregated as is the case in broad bands. Second, its 30-m spatial resolution renders it more suitable than ocean color sensors (e.g., Sentinel-3 OLCI) for monitoring medium- to small-sized inland lakes and reservoirs.

Owing to the recent launch of the ZY1-02D satellite, studies on water quality monitoring based on its data are limited [21], [22]. Considering the abundant bands provided by AHSI, a classification-based SAM is established here to provide accurate $Z_{sd}$ estimates using the AHSI images from ZY1-02D for both turbid and clear inland waters. This method can also be applied to any sensor with bands centered at 443, 490, 532, 555, 620, 665, and 708 nm. In this study, a simple image preprocessing scheme is presented for remote sensing reflectance ($R_{rs}$) image retrieval from the ZY1-02D AHSI data. Further, by comparing the proposed method with the existing empirical/semianalytical models, the effectiveness and potential for water clarity estimations, using the ZY1-02D AHSI imagery, are also tested.

## II. Data

### A. $R_{rs}$ and $Z_{sd}$ Field Measurements

We conducted 11 field experiments on nine inland lakes and reservoirs in China from May 2019 to October 2020. Among the water bodies, the Panjiakou, Danjiangkou, and Xiaolangdi reservoirs have relatively clear waters, whereas the Baiyangdian Lake and Taihu Lake and the Yuqiao, Guanting, Daheiting, and Baiguishan reservoirs contain turbid waters. In the field, the remote sensing reflectance, $R_{rs}$, and $Z_{sd}$, were measured at 200 sites. Table I presents the measurement time and $Z_{sd}$ ranges.

At each sampling site, $Z_{sd}$ was measured using a standard 30-cm-diameter Secchi disk. In addition, the water-leaving radiance ($L_u$), the radiance of a standard gray board ($L_b$), and the skylight radiance ($L_{sky}$) were measured using the above-water method [23] with a FieldSpec analytical spectral device (ASD) spectrometer. The remote sensing reflectance was then calculated as follows [24]:

$$R_{rs}^{m}(\lambda) = \frac{L_u(\lambda) - \rho_{sky}(\lambda) L_{sky}(\lambda)}{\pi L_b(\lambda)/\rho_b(\lambda)} - \Delta \tag{1}$$

where $R_{rs}^{m}$ is the *in situ* measured $R_{rs}$, $\lambda$ represents the wavelength, and $\rho_b$ and $\rho_{sky}$ are the reflectance of the standard gray board and skylight, respectively, where $\Delta$ ($sr^{-1}$) accounts for the skylight from other directions that is also reflected into the sensor's field of view from the rough water surface. Specifically, $\rho_b$ was calibrated in the laboratory, while $\rho_{sky}$ values can be obtained from the lookup table in [25]. As described in [24], $\Delta$ is determined by setting the average of $R_{rs}^{m}$ (1600–1650 nm) to 0. Fig. 2 displays the *in situ* measured $R_{rs}^{m}$ spectra of all of the sampling sites. Fig. 3 shows the distribution of the sampling sites.

### B. Concurrent ZY1-02D Image Acquisition

The ZY1-02D satellite hyperspectral images were selected in accordance with satellite overpasses within $\pm 1$ day of the field measurements, considering stable meteorological conditions (i.e., no rain with a wind speed of $<5$ m/s) during the satellite image acquisition and *in situ* data collection (see Table I). Specifically, the images of the Taihu Lake and the Danjiangkou and Guanting reservoirs were acquired on the same day as the field surveys, whereas the image for the Xiaolangdi reservoir was acquired a day before the *in situ* measurements. As Xiaolangdi reservoir is located in the

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TABLE I

LOCATION, DATES, NUMBER OF SAMPLING SITES ($n$), AND $Z_{sd}$ RANGES OF 11 FIELD EXPERIMENTS, WITH CONCURRENT ZY1-02D ACQUISITION DATES

| No. | Waterbody Name | Lat. (N) | Long. (E) | Experiment Date | N | $Z_{sd}$ ranges (m) | ZY1-02D Acquisition Date |
|---|---|---|---|---|---|---|---|
| 1 | Taihu Lake | 30.22 | 121.18 | May 1, 2019 | 9 | 0.4–1.6 | |
| 2 | Baiyangdian Lake | 38.82 | 116.01 | May 21–22, 2019 | 29 | 0.5–1.7 | |
| 3 | Guanting Reservoir | 40.35 | 115.73 | May 22, 2019 | 18 | 0.3–2.2 | |
| 4 | Panjiakou Reservoir | 40.43 | 118.29 | Sep. 24, 2019 | 25 | 1.2–4.5 | |
| 5 | Daheiting Reservoir | 40.28 | 118.31 | Sep. 25, 2019 | 12 | 0.8–2.1 | |
| 6 | Yuqiao Reservoir | 40.04 | 117.54 | Oct. 8, 2019 | 19 | 0.4–1.0 | |
| 7 | Baiguishan Reservoir | 33.74 | 113.17 | Sep. 4, 2020 | 18 | 0.8–1.8 | |
| 8 | Guanting Reservoir | 40.35 | 115.73 | Aug. 13, 2020 | 18 | 0.4–0.7 | Aug. 13, 2020 |
| 9 | Taihu Lake | 30.22 | 121.18 | Sep. 6, 2020 | 18 | 0.2–0.7 | Sep. 6, 2020 |
| 10 | Danjiangkou Reservoir | 32.72 | 111.57 | Sep. 2, 2020 | 17 | 2.9–6.4 | Sep. 2, 2020 |
| 11 | Xiaolangdi Reservoir | 34.95 | 112.28 | Oct. 22, 2020 | 17 | 1.3–3.2 | Oct. 21, 2020 |

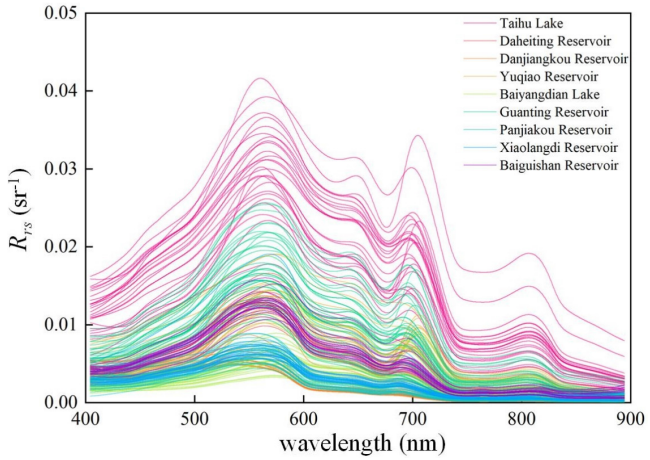



Fig. 2. *In situ* measured $R_{rs}^{m}$ spectra of all of the sampling sites from nine inland lakes and reservoirs.

National Nature Reserves of the Henan Yellow River Wetland, human activities are highly restricted for the protection of the wetland ecosystem [26]. Besides, the Xiaolangdi reservoir has deeper waters (>40 m). Thus, the water quality of Xiaolangdi reservoir is relatively stable and unlikely to be affected by human activities. All images were acquired at approximately 11:00 local time and had a <10% cloud coverage. Matching pairs of the image-derived and *in situ* remote sensing reflectance were selected according to the following criteria [27], [28]: a 3 × 3 pixel window centered over the *in situ* sampling site was selected on the image and the central pixel was considered a match if the coefficient of variation (i.e., standard deviation/mean) was <0.40 in the window [28]. All pixels corresponding to the 70 matching sampling sites (see Table I) satisfied this condition.

## III. METHOD

### A. ZY1-02D Band $R_{rs}$ Simulations

The *in situ* measured $R_{rs}^{m}$ data were converted into that of the ZY1-02D bands using the following spectral response function (SRF):

$$R_{rs}^{m}(B_i) = \frac{\int_{\lambda_{min}}^{\lambda_{max}} R_{rs}^{m}(\lambda) f_i(\lambda) d\lambda}{\int_{\lambda_{min}}^{\lambda_{max}} f_i(\lambda) d\lambda} \quad (2)$$

where $B_i$ is the $i$th ZY1-02D band, $f_i(\lambda)$ is the SRF of the $i$th band, and $\lambda_{min}$ and $\lambda_{max}$ are the lower and upper wavelengths of $f_i(\lambda)$, respectively. Here, $f_i(\lambda)$ was assumed to be the Gaussian function, which is typically used for hyperspectral data [29], [30]. The Gaussian SRF is as follows:

$$f_i(\lambda) = \exp\left[-4\ln(2)\left(\frac{\lambda - \lambda_i}{\Delta\lambda_i}\right)^2\right] \quad (3)$$

where $\lambda_i$ and $\Delta\lambda_i$ are the center wavelength and the full-width at half-maximum (FWHM) of the $i$th band, respectively. FWHMs of the VNIR and SWIR bands of ZY1-02D were 8.76 and 16.26 nm, respectively. The simulated AHSI band $R_{rs}$ was then used to develop the $Z_{sd}$ retrieval method with their paired *in situ* measured $Z_{sd}$, including threshold determination for the Hybrid QAA-based SAM and empirical model tuning in Section III-B.

### B. $Z_{sd}$ Model Development Using in Situ Dataset

*1) Hybrid QAA-Based SAM:* The SAM proposed by Lee *et al.* [8] involves the following three major steps for $Z_{sd}$ retrieval: 1) estimating the inherent optical properties (IOPs), i.e., the total absorption coefficient $[a(\lambda)]$ and backscattering coefficient $[b_b(\lambda)]$ using the version 5 QAA ($QAA_{v5}$) [31]; 2) determining the diffuse attenuation coefficient ($K_d$, $m^{-1}$) based on $a(\lambda)$ and $b_b(\lambda)$; and 3) retrieving $Z_{sd}$ using $K_d$ and $R_{rs}$. The $a(\lambda)$ and $b_b(\lambda)$ estimation in the first step is vital for $Z_{sd}$ retrieval because the accuracy of the derived $K_d$ and $Z_{sd}$ is highly affected by the IOPs estimated using the QAA [32], [33]. Nevertheless, $QAA_{v5}$ has limitations in providing accurate IOP estimates for turbid and eutrophic inland waters [14]. As $a(555)$ may not be dominated by $a_w(555)$ in highly turbid inland waters, using 555 nm as the reference wavelength in $QAA_{v5}$ and the empirical equation for the nonwater absorption constituent estimations could be unreliable. Therefore, various QAA versions [12]–[14], [33] were developed to accommodate different water types by introducing longer reference wavelengths and distinct empirical parameters. Considering the limitation of $QAA_{v5}$, we included another QAA, which was developed for turbid and hypereutrophic waters by Mishra *et al.* [14] (hereafter referred to as $QAA_{m14}$). The $QAA_{m14}$ was chosen because it has been utilized in many studies [18], [33], [34] that have demonstrated its robustness for retrieving the IOPs of turbid waters.

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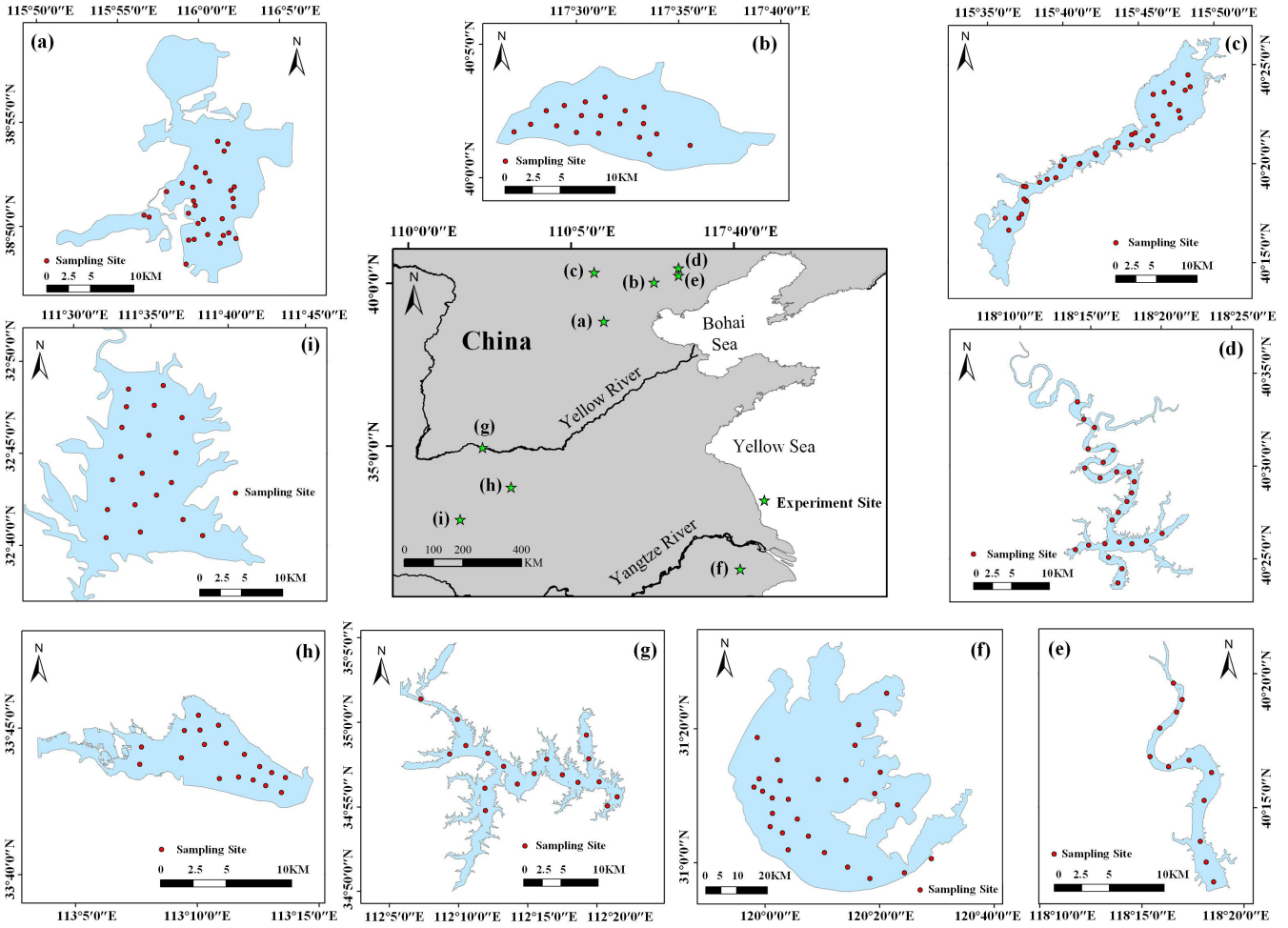



Fig. 3. Distribution of the sampling sites for nine inland lakes and reservoirs in Eastern China, including the (a) Baiyangdian Lake, (b) Yuqiao reservoir, (c) Guanting reservoir, (d) Panjiakou reservoir, (e) Daheiting reservoir, (f) Taihu Lake, (g) Xiaolangdi reservoir, (h) Baiguishan reservoir, and (i) Danjiangkou reservoir. Measurements were carried out at 200 sites during 11 field experiments between 2019 and 2020. The dates and details of the sampling sites for each experiment are listed in Table I.

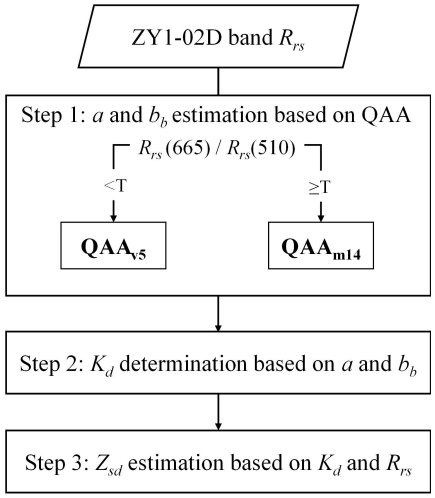



Fig. 4. Flowchart of the classification-based ZY1-02D $Z_{sd}$ retrieval method involving a hybrid QAA.

Fig. 4 details the process of the $Z_{sd}$ retrieval method proposed in this study.

First, we established a hybrid QAA method that employs the $QAA_{v5}$ and $QAA_{m14}$, which select different reference wavelengths, $\lambda_0$, to calculate $a(\lambda_0)$ in clear to moderately turbid and highly turbid water types, respectively. In this hybrid QAA, the water type is determined according to the $R_{rs}$ ratio of the red to blue band (red/blue). As the signal in the red band usually increases with a decrease in the water clarity, this ratio is generally low for relatively clear water and high for very turbid water [35]. As shown in Fig. 4, a threshold, $T$, determined via trial and error using the *in situ* dataset, was introduced for the classification. The ZY1-02D band-averaged *in situ* $R_{rs}$ centered at 665 and 510 nm was used to represent the red and blue bands, respectively, which were employed for calculating the red/blue ratio.

In Fig. 5, $Z_{sd}$ decreases rapidly as $R_{rs}(665)/R_{rs}(510)$ increases from 0.2 to ~0.6 while exhibiting insensitivity to higher $R_{rs}(665)/R_{rs}(510)$ values. We set the threshold (i.e., $T$ in Fig. 4) to 0.5–0.6 at an interval of 0.01. The overall estimation accuracy was highest for $Z_{sd}$ values at the 200 sampling sites when $T$ was set to 0.55. Therefore, clear to moderately turbid water and highly turbid water were separated using an $R_{rs}(665)/R_{rs}(510)$ value of 0.55. Consequently, the $QAA_{v5}$ was chosen for calculating the IOPs when $R_{rs}(665)/R_{rs}(510)$ was <0.55, while the $QAA_{m14}$ was employed otherwise.

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TABLE II
STEPS FOR DERIVING THE ABSORPTION AND BACKSCATTERING COEFFICIENTS USING THE HYBRID QAA

| Step | Property[1] | Equation | |
|---|---|---|---|
| 1 | $r_{rs}$ | $= R_{rs}/(0.52 + 1.7R_{rs})$ | |
| 2 | $u(\lambda)$ | $= \frac{-g_0+\sqrt{g_0^2+4g_1 r_{rs}(\lambda)}}{2g_1}$, where $g_0 = 0.089$ $g_1 = 0.125$ | |
| 3 | $a(\lambda_0)$ | $= a_w(\lambda_0) + 10^{a+b*\chi+c*\chi^2}$ | |
| | | **IF** $\frac{R_{rs}(665)}{R_{rs}(510)} < 0.55$<br>$\lambda_0 = 555nm$<br>$a = -1.146, b = -1.366, c = -0.469$<br>$\chi = \log(\frac{r_{rs}(443)+r_{rs}(490)}{r_{rs}(\lambda_0)+5\frac{r_{rs}(665)}{r_{rs}(490)}r_{rs}(665)})$ | **(else)**<br>$\lambda_0 = 708nm$<br>$a = -0.7153, b = -2.054, c = -1.047$<br>$\chi = \log(\frac{0.001r_{rs}(443)+r_{rs}(620)}{r_{rs}(\lambda_0)+0.005\frac{r_{rs}(620)}{r_{rs}(443)}r_{rs}(620)})$ |
| 4 | $b_{bp}(\lambda_0)$ | $= \frac{u(\lambda_0)a(\lambda_0)}{1-u(\lambda_0)} - b_{bw}(\lambda_0)$ | |
| 5 | $\eta$ | $= 2.0\{1 - 1.2exp[-0.9\frac{r_{rs}(443)}{r_{rs}(555)}]\}$ | |
| 6 | $b_{bp}(\lambda)$ | $= b_{bp}(\lambda_0)(\frac{\lambda_0}{\lambda})^{\eta}$ | |
| 7 | $a(\lambda)$ | $= \frac{[1-u(\lambda)][b_{bp}(\lambda)+b_{bw}(\lambda)]}{u(\lambda)}$ | |

* [1]$r_{rs}$ = below-surface remote-sensing reflectance; $u(\lambda)$ = ratio of the backscattering coefficient to the sum of the absorption and backscattering coefficients ($b_b/(a+b_b)$); $\lambda_0$ = reference wavelength; $b_{bp}$ = backscattering coefficient of suspended particles; $\eta$ = spectral power of the particle scattering coefficient.

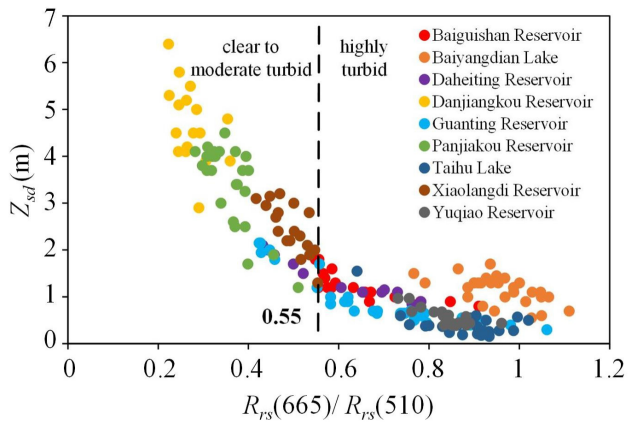



Fig. 5. Scatterplot of $Z_{sd}$ versus $R_{rs}(665)/R_{rs}(510)$ using *in situ* data collected at 200 sampling sites from nine lakes and reservoirs.

Table II presents the calculation steps using the $QAA_{v5}$ and $QAA_{m14}$.

Second, $K_d$ was then derived using the estimated $a(\lambda)$ and $b_b(\lambda)$ as follows [36]:

$$K_d(\lambda) = (1 + m_0 \times \theta_s)a(\lambda) + \left(1 - \gamma \frac{b_{bw}(\lambda)}{b_b(\lambda)}\right) \times m_1 \times \left(1 - m_2 \times e^{-m_3 \times a(\lambda)}\right) b_b(\lambda) \quad (4)$$

where $\theta_s$ is the solar zenith angle in degrees, $b_{bw}(\lambda)$ is the backscattering coefficient of pure water, and $m_{0-3}$ and $\gamma$ are constants (i.e., 0.005, 4.26, 0.52, 10.8, and 0.265) [31].

Finally, based on the estimated $K_d$ and $R_{rs}$, $Z_{sd}$ was obtained as follows [8]:

$$Z_{sd} = \frac{1}{2.5\text{Min}\left(K_d^{tr}\right)} \ln\left(\frac{\left|0.14 - R_{rs}^{tr}\right|}{0.013}\right) \quad (5)$$

where $\text{Min}(K_d^{tr})$ refers to the minimum diffuse attenuation coefficient at the water's transparent window in the visible spectral range (410–665 nm), and it was determined from values at 443, 490, 532, 555, and 665 nm; $R_{rs}^{tr}$ denotes the corresponding remote sensing reflectance at this wavelength.

*2) Empirical Model Tuning:* To assess the performance of the classification-based SAM, several widely used empirical models were optimized and applied to the ZY1-02D data. The parameters in the existing empirical models were tuned to fit the *in situ* dataset presented in Table I using the 130 *in situ* samples without concurrent ZY1-02D images employed for calibration. The remaining 70 samples were used for validation. The empirical model that exhibited the best performance on both the calibration and validation datasets was then selected to test the ZY1-02D imagery. The image-derived $Z_{sd}$ was finally compared with those derived from the classification-based SAM.

*3) Accuracy Assessment:* Three statistical parameters, including the coefficient of determination ($R^2$), average unbiased relative error (AURE), and root mean square error (RMSE), were used to assess the results of the classification-based SAM and the calibrated empirical models. AURE and RMSE are defined as follows:

$$\text{AURE} = \frac{1}{N}\sum_{i=1}^{N} \frac{|Y_i - X_i|}{0.5(Y_i + X_i)} \times 100\% \quad (6)$$

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$$\mathrm{RMSE} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(Y_i - X_i)^2} \tag{7}$$

where $Y_i$ and $X_i$ represent the derived and field-measured $Z_{sd}$ of the $i$th sample, respectively. In (6), $0.5(Y_i + X_i)$ was used as the denominator instead of $X_i$ to alleviate the interference associated with spatial and temporal scale differences between the *in situ* and image matchups, which are often employed in evaluations of water constituent estimation models [37], [38].

### C. $Z_{sd}$ Retrieval Based on ZY1-02D Images

To retrieve $Z_{sd}$ from the ZY1-02D images, we performed atmospheric correction to retrieve $R_{rs}$ from the original AHSI images, followed by water area extraction. The $Z_{sd}$ values were then estimated based on the image-derived $R_{rs}$ using the classification-based SAM and optimized empirical model. For the $K_d$ calculation, the solar zenith angle was set to the value in the central pixel of each image. This did not induce large errors in the $Z_{sd}$ estimation as the swath of ZY1-02D images was only 60 km. Finally, an accuracy analysis was conducted for the ZY1-02D image-retrieved $Z_{sd}$ for both methods via a comparison with the field-measured $Z_{sd}$ of the 70 matched sampling sites.

*1) ZY1-02D Image Preprocessing:* First, the original AHSI images were corrected for surface reflectance using the ENVI software. Specifically, the VNIR and SWIR band data were merged and converted to top-of-atmosphere radiance data using the radiometric calibration coefficients, which were determined through vicarious calibration [39] performed at the Baotou Test Site [40]. The surface reflectance images were acquired using the fast line-of-sight atmospheric analysis of spectral hypercubes (FLAASHs) module [41] in ENVI. As the Guanting, Danjiangkou, and Xiaolangdi reservoirs are located in rural areas, we chose the rural aerosol model for these regions. For Taihu Lake, which is surrounded by urban areas, we used both the urban and rural models in the FLAASH module, finding that the retrieved surface reflectance using the rural model was more accurate when compared with that of the *in situ* measurements. Therefore, the rural model was used for all four study areas. We used the two-band Kaufman–Tanre method [42] for the aerosol retrieval. In addition, the 940-nm band was chosen for water vapor retrieval.

Based on the surface reflectance image, the normalized water index (NDWI) image was calculated [43]. Water areas were then automatically segmented using the modified histogram bimodal method (MHBM) [44]. The algal bloom and water grass areas present in the Taihu Lake image were excluded using the threshold segmentation of the floating algae index (FAI) [45].

Removal of the reflected skylight from the surface reflectance is necessary for accurate $R_{rs}$ retrieval. However, conducting accurate skylight removal without a concurrent measurement for the atmospheric conditions is difficult. Despite the spectral variation in the skylight signals, some previous studies [28], [46] assumed that the skylight to be the same for all bands and removed it using the NIR and SWIR bands. This method has yielded satisfactory $R_{rs}$ estimates for moderate resolution imaging spectroradiometer (MODIS) data, with successful utilization for large-scale inland water monitoring [47]. As no proper skylight removal method has been proposed for ZY1-02D data, we also adopted this method for the $R_{rs}$ calculation of ZY1-02D as follows [46]:

$$R_{rs}^{c}(\lambda) = \frac{\rho(\lambda) - \min(\rho_{NIR}, \rho_{SWIR})}{\pi} \tag{8}$$

where $R_{rs}^{c}(\lambda)$ is the corrected $R_{rs}$ and $\min(\rho_{NIR}, \rho_{SWIR})$ is the minimum reflectance value of the NIR and SWIR bands. To minimize the single band noise interference, we averaged the surface reflectance values of the ZY1-02D bands between 720 and 730 and 1530 and 1630 nm to calculate $\rho_{NIR}$ and $\rho_{SWIR}$, respectively.

*2) Atmospheric Correction Performance Evaluation:* Accurate retrieval of the image-derived remote sensing reflectance is a prerequisite for reliable $Z_{sd}$ retrieval. Hence, we first evaluated the atmospheric correction performance of the ZY1-02D imagery by comparing $R_{rs}^{c}$ to its concurrent *in situ* simulated ZY1-02D-band $R_{rs}^{m}(B_i)$. The $R_{rs}^{c}$ values of the 70 matched pixels were evaluated based on the accuracy of $R_{rs}^{c}$ in the ZY1-02D bands and the spectral shape similarity between $R_{rs}^{c}$ and $R_{rs}^{m}$ for each sampling site. Specifically, the accuracy of the $R_{rs}^{c}$ data was assessed using the averaged $R_{rs}^{c}/R_{rs}^{m}$ ratio, the mean relative error (MRE) of $R_{rs}^{c}$, and $R^2$; the spectral shape similarity was evaluated using the correlation coefficient, $R$, between the ZY1-02D-band $R_{rs}^{c}$ and the *in situ* simulated $R_{rs}^{m}$ in the 443–720 nm spectral range.

*3) ZY1-02D-Retrieved $Z_{sd}$ Accuracy Assessment:* Based on the image-derived $R_{rs}^{c}$, $Z_{sd}$ was derived using our classification-based SAM and the optimal empirical model. The accuracy analysis of the image-derived $Z_{sd}$ at the 70 matched points was conducted using AURE and RMSE.

## IV. Results

### A. In Situ Data-Based $Z_{sd}$ Model Development

*1) $Z_{sd}$ Estimation Based on SAM:* Using 0.55 as the $R_{rs}(665)/R_{rs}(510)$ threshold, $Z_{sd}$ was derived based on the *in situ* simulated ZY1-02D-band $R_{rs}^{m}(B_i)$ using the classification-based SAM. The $Z_{sd}$ values were also estimated with the SAM using only the $\mathrm{QAA_{v5}}$ or $\mathrm{QAA_{m14}}$. For comparison, the $Z_{sd}$ values were also estimated based on SAM with $\mathrm{QAA_{v6}}$ [48], in which the calculation of $a$ and $b_{bp}$ was different from $\mathrm{QAA_{v5}}$ for $R_{rs}(670) \geq 0.0015\ \mathrm{sr}^{-1}$. SAM models based on other QAAs developed for inland waters were also tested; these QAA algorithms were labeled as $\mathrm{QAA_{L09}}$ [12], $\mathrm{QAA_{Y13}}$ [13], and $\mathrm{QAA_{w16}}$ [49]. For comparison with the performance of the SAM when applied to the Landsat-8 band $R_{rs}$, $Z_{sd}$ retrieval based on [50] (labeled as QAA-L8) and QAA-RGB [51] was also included. Table III lists the corresponding accuracies for $Z_{sd}$ estimated using various SAMs. Based on Table III, the classification-based SAM based on the hybrid QAA yielded better $Z_{sd}$ estimates than with any QAA alone, as the former estimated $Z_{sd}$ had a higher $R^2$, lower AURE, and lower RMSE based on the 200 sampling sites.

*2) $Z_{sd}$ Estimation Based on the Empirical Models:* We calibrated several widely used and newly developed empirical models [5], [6], [52]–[59] using the *in situ* simulated

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TABLE III

AURE, RMSE, AND $R^2$ OF THE ESTIMATED $Z_{sd}$ BASED ON *In Situ* ZY1-02D BAND $R_{rs}$ FROM 200 SAMPLING SITES USING VARIOUS QAA ALGORITHMS. $N$ REPRESENTS THE NUMBER OF BANDS APPLIED IN THE CORRESPONDING QAA. BEST RESULTS ARE IN BOLD

| AURE | RMSE (m) | $R^2$ | Reference | N |
|---|---|---|---|---|
| 35.9% | 0.48 | 0.93 | $QAA_{v5}$ [31] | 5 |
| 37.8% | 0.60 | 0.87 | $QAA_{v6}$ [48] | 5 |
| 30.6% | 0.86 | 0.87 | $QAA_{L09}$ [12] | 7 |
| 39.1% | 1.09 | 0.59 | $QAA_{Y13}$ [13] | 7 |
| 21.2% | 0.54 | 0.86 | $QAA_{m14}$ [14] | 7 |
| 21.5% | 0.66 | 0.87 | $QAA_{w16}$ [49] | 7 |
| 27.7% | 0.47 | 0.91 | QAA-L8 [50] | 4 |
| 38.9% | 0.63 | 0.93 | QAA-RGB [51] | 3 |
| **17.6%** | **0.32** | **0.95** | Hybrid QAA | 7 |

TABLE IV

NUMBER OF SAMPLES IN THE CALIBRATION AND VALIDATION DATASET FOR DIFFERENT WATER CLASSES

| | The number of samples | |
|---|---|---|
| | Calibration | Validation |
| Class 1 ($R_{rs}(665)/R_{rs}(510)<0.55$) | 37 | 39 |
| Class 2 ($R_{rs}(665)/R_{rs}(510)\geq 0.55$) | 93 | 31 |
| Total | 130 | 70 |

ZY1-02D-band $R_{rs}$ of the 130 sampling points without concurrent ZY1-02D images in the *in situ* dataset. For comparison with the classification-based SAM, empirical models were separately built for the sampling sites in different classes. To achieve this, 130 and 70 samples in the calibration and validation datasets were divided into two groups (see Table IV).

Considering the band specifications of the AHSI, we utilized the bands centered at 665, 555, and 510 nm as the red, green, and blue bands, respectively. Tables V and VI present the corresponding optimized empirical equations for both classes. Using these equations, the $Z_{sd}$ values were estimated for the 70 validation samples (i.e., 39 samples in Class 1 and 31 samples in Class 2). Table VII presents the corresponding estimation errors.

As listed in Tables V and VI, the empirical model based on $R_{rs}(555)$ and $R_{rs}(510)/R_{rs}(665)$ exhibited the highest $R^2$ for the calibration dataset. This model also produced the highest $R^2$ for the validation dataset, with a relatively low AURE and RMSE (see Table VII). Therefore, this model was chosen for application to the ZY1-02D images for comparison.

*3) Semianalytical and Empirical Methods Comparison:* To enable a comparison of classification-based SAM and empirical methods, the AURE and RMSE for $Z_{sd}$ retrieved using the classification-based SAM were also calculated for the validation dataset (i.e., 70 samples) used to develop the empirical models. The AURE and RMSE of $Z_{sd}$ derived from the classification-based method were 19.7% and 0.36 m,

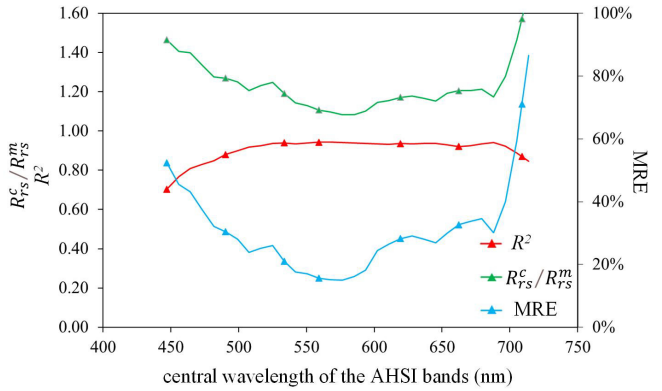



Fig. 6. Accuracy analysis of $R_{rs}^c$ in the ZY1-02D bands of 443–720 nm. Triangle marker denotes the bands applied in the hybrid QAA.

respectively, with an $R^2$ of 0.97. Compared with the results in Table VII, the classification-based SAM displayed an overall better performance, involving lower AURE and RSME and higher $R^2$ relative to the calibrated empirical models. According to the *in situ* results, the classification-based SAM exhibited flexibility for application in highly turbid to clear waters and provided superior $Z_{sd}$ estimates compared with the SAM integrated with only one QAA and the calibrated empirical models.

Further, as the empirical methods used two to four bands and SAMs based on QAA-L8 and QAA-RGB required four and three operational land imager (OLI) bands, respectively, the *in situ* experimental results showed that the utilization of more ($N = 7$) and narrow band $R_{rs}$ aided in the estimation of $Z_{sd}$ with a higher accuracy.

### *B. $Z_{sd}$ Results Based on ZY1-02D Images*

To provide a more precise analysis of the AHSI-retrieved $Z_{sd}$ results, only 53 matchups were used for the accuracy assessment in this section. Considering the one-day difference between the *in situ* measurement and AHSI image acquisition, 17 sampling points in the Xiaolangdi reservoir were excluded.

*1) Image-Derived $R_{rs}$ Accuracy Assessment:* According to the atmospheric correction performance evaluation process described in Section III–C, the 53 matched image-derived $R_{rs}^c$ were compared with the corresponding *in situ* $R_{rs}^m$. The $R_{rs}^c$ accuracy was evaluated, as shown in Fig. 6. Table VIII presents the detailed values of the accuracy indicators used in the bands of the hybrid QAA.

From the MRE curve in Fig. 6, we observed that the image-derived $R_{rs}^c$ was accurate in most bands, except for bands centered at <500 and >700 nm. For wavelengths <500 nm, the increased error may have been due to inadequate Rayleigh scattering correction in the atmospheric correction for the blue bands. In bands centered at >700 nm, $R_{rs}^c$ with substantial errors mainly existed in matchups for the Danjiangkou and Xiaolangdi reservoirs, in which the water is clear and the image spectra were therefore more affected by noise. As listed in Table VIII, in the bands centered at 490, 532, 555, 620, and 665 nm, the averaged $R_{rs}^c/R_{rs}^m$ of the 53 matchups were all approximately 1, with an $R^2$ of $\geq$0.88.

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TABLE V

CALIBRATED EMPIRICAL MODELS FOR CLASS 1 BASED ON THE *In Situ* DATA FROM 37 SAMPLING SITES. MODEL WITH HIGHEST $R^2$ IS IN BOLD

| Reference | Band or spectral index | Calibrated $Z_{sd}$ model | Calibration dataset $R^2$ |
|---|---|---|---|
| [5] | $x = R_{rs}(555)$ | $6.74e^{-105.4x}$ | 0.82 |
| [6] | $x = R_{rs}(665)/R_{rs}(510)$ | $14.51e^{-3.33x}$ | 0.84 |
| [52] | $x = R_{rs}(665)/R_{rs}(555)$ | $23.94e^{-6.61x}$ | 0.59 |
| [53] | $x = R_{rs}(510)/R_{rs}(665)$ | $0.816x^{1.70}$ | 0.81 |
| [54] | $x_1 = R_{rs}(510)\ x_2 = R_{rs}(665)$ | $e^{1.38+145.59x_1-401.22x_2}$ | 0.87 |
| [55] | $x_1 = R_{rs}(555)\ x_2 = R_{rs}(665)/R_{rs}(510)$ | $e^{2.44-49.56x_1-2.01x_2}$ | **0.91** |
| [56] | $x_1 = R_{rs}(665)/R_{rs}(555)$<br>$x_2 = R_{rs}(510)/R_{rs}(665)$ | $-2.72 + 2.12x_1 + 2.38x_2$ | 0.83 |
| [57] | $x = (R_{rs}(520) - R_{rs}(781))/(R_{rs}(700) - R_{rs}(781))$ | $-1.89 + 1.66x$ | 0.57 |
| [58] | $x = (R_{rs}(555) + (R_{rs}(665))/2$ | $6.33e^{-146.2x}$ | 0.83 |
| [59] | $x = R_{rs}(665)$ | $5.40e^{-232.0x}$ | 0.82 |

TABLE VI

CALIBRATED EMPIRICAL MODELS FOR CLASS 2 BASED ON THE *In Situ* DATA FROM 93 SAMPLING SITES. MODEL WITH HIGHEST $R^2$ IS IN BOLD

| Reference | Band or spectral index | Calibrated $Z_{sd}$ model | Calibration dataset $R^2$ |
|---|---|---|---|
| [5] | $x = R_{rs}(555)$ | $0.121x^{-0.437}$ | 0.32 |
| [6] | $x = R_{rs}(665)/R_{rs}(510)$ | $0.717x^{-0.830}$ | 0.14 |
| [52] | $x = R_{rs}(665)/R_{rs}(555)$ | $0.60x^{-0.560}$ | 0.07 |
| [53] | $x = R_{rs}(510)/R_{rs}(665)$ | $0.373e^{0.66x}$ | 0.16 |
| [54] | $x_1 = R_{rs}(510)\ x_2 = R_{rs}(665)$ | $e^{0.086+59.976x_1-107.729x_2}$ | 0.37 |
| [55] | $x_1 = R_{rs}(555)\ x_2 = R_{rs}(665)/R_{rs}(510)$ | $e^{1.20-33.169x_1-1.124x_2}$ | **0.44** |
| [56] | $x_1 = R_{rs}(665)/R_{rs}(555)$<br>$x_2 = R_{rs}(510)/R_{rs}(665)$ | $0.97 - 1.262x_1 + 0.982x_2$ | 0.16 |
| [57] | $x = (R_{rs}(520) - R_{rs}(781))/(R_{rs}(700) - R_{rs}(781))$ | $0.690x^{0.677}$ | 0.30 |
| [58] | $x = (R_{rs}(555) + (R_{rs}(665))/2$ | $0.094x^{-0.466}$ | 0.36 |
| [59] | $x = R_{rs}(665)$ | $0.069x^{-0.491}$ | 0.44 |

TABLE VII

ACCURACIES OF THE ESTIMATED $Z_{sd}$ USING DIFFERENT EMPIRICAL MODELS BASED ON *In Situ* DATA. BEST RESULTS ARE IN BOLD

| Reference | Validation dataset | | |
|---|---|---|---|
| | AURE | RMSE(m) | $R^2$ |
| [5] | 43.9% | 0.85 | 0.90 |
| [6] | 55.5% | 1.12 | 0.92 |
| [52] | 53.6% | 0.98 | 0.86 |
| [53] | 60.6% | 1.84 | 0.91 |
| [54] | 44.2% | 0.91 | 0.91 |
| [55] | 39.7% | **0.73** | **0.95** |
| [56] | 61.5% | 1.41 | 0.92 |
| [57] | 67.5% | 2.45 | 0.92 |
| [58] | 42.1% | 0.81 | 0.91 |
| [59] | **39.4%** | 0.77 | 0.92 |

TABLE VIII

AVERAGES OF THE $R^c_{rs}/R^m_{rs}$, MRE, AND $R^2$ OF THE 53 MATCHUPS IN ALL BANDS APPLIED IN THE HYBRID QAA FOR THE IMAGE-CORRECTED $R^c_{rs}$

| Band (nm) | $R^c_{rs}/R^m_{rs}$ | MRE | $R^2$ |
|---|---|---|---|
| 443 | 1.46 | 52.3% | 0.70 |
| 490 | 1.27 | 30.4% | 0.88 |
| 532 | 1.19 | 21.0% | 0.94 |
| 555 | 1.11 | 15.6% | 0.94 |
| 620 | 1.17 | 28.2% | 0.94 |
| 665 | 1.21 | 32.7% | 0.92 |
| 708 | 1.57 | 71.2% | 0.87 |

Further, the image-derived $R^c_{rs}$ exhibited spectral shapes consistent with its *in situ* matched $R^m_{rs}$ (see Fig. 7).

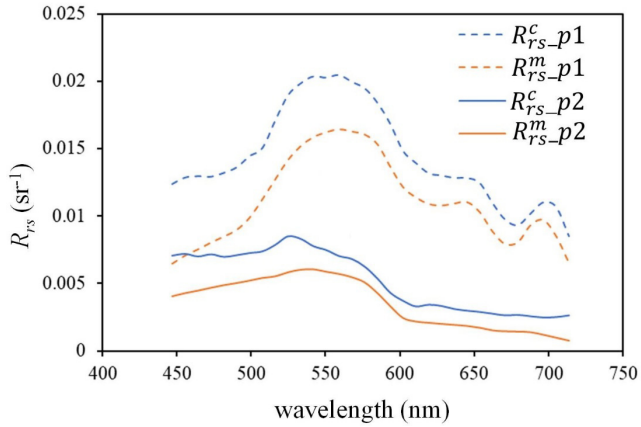



Fig. 7. $R^c_{rs}$ compared to its corresponding *in situ* $R^m_{rs}$; $p1$ and $p2$ are the two representative matched points shown as examples.

The correlation coefficients of the 53 matchups varied between 0.835 and 0.992, with a mean value of 0.931, indicating a high similarity between the spectral $R^c_{rs}$ and $R^m_{rs}$.

*2) Image-Derived $Z_{sd}$ Accuracy Assessment:* Using the classification-based SAM integrated with the hybrid QAA, $Z_{sd}$ was retrieved from the ZY1-02D AHSI images. Fig. 8 plots the image-estimated $Z_{sd}$ values of the 53 matchups with the *in situ* measured $Z_{sd}$. Table IX presents the accuracy assessment data for the values estimated using the classification-based SAM and the empirical model.

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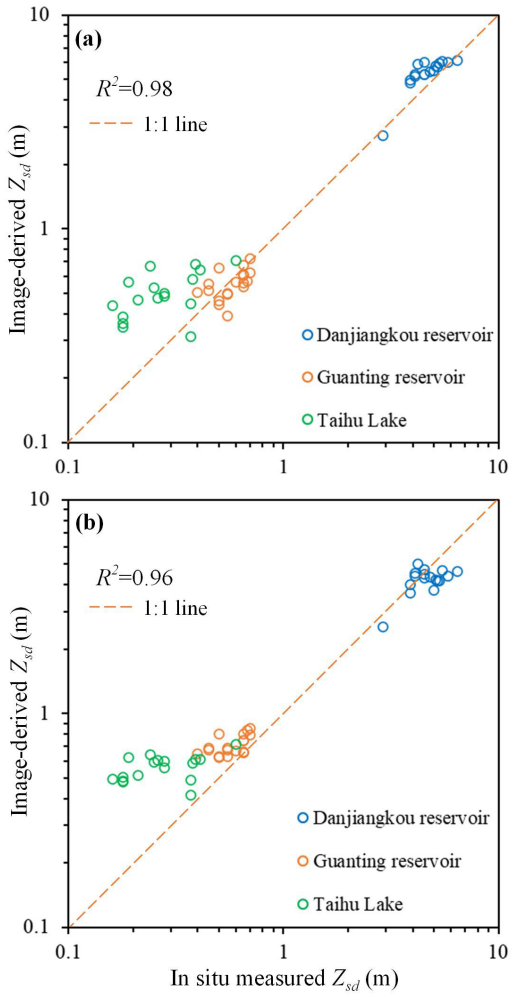



Fig. 8. Scatterplots of $Z_{sd}$ estimated from the ZY1-02D image using the classification-based SAM, as well as $R_{rs}(555)$- and $R_{rs}(510)$/$R_{rs}(665)$-based empirical model versus the *in situ* measured $Z_{sd}$ for the 53 matchups. (a) Classification-based SAM mode. (b) Empirical model.

TABLE IX

ACCURACY ASSESSMENT FOR THE ESTIMATED $Z_{sd}$ OF THE 53 MATCHUPS FROM THE ZY1-02D HYPERSPECTRAL IMAGES BASED ON THE CLASSIFICATION-BASED SAM AND THE BLUE/RED-BASED EMPIRICAL MODEL. BEST RESULTS ARE IN BOLD

| $Z_{sd}$ Range (m) | N | Classification-based SAM | | Empirical model | |
|---|---|---|---|---|---|
| | | AURE | RMSE (m) | AURE | RMSE |
| 0–3 | 37 | **34.9%** | **0.18** | 43.0% | 0.23 |
| > 3 | 16 | 16.1% | 0.90 | **14.5%** | **0.86** |
| Total | 53 | **29.1%** | 0.52 | 34.2% | **0.51** |

The data in Fig. 8 reveal a consistency between the predicted $Z_{sd}$ using the classification-based SAM and the *in situ* measurements, with an $R^2$ of 0.98. The image-derived $Z_{sd}$ using the empirical model produced an $R^2$ of 0.96. Nevertheless, the $Z_{sd}$ estimates using the classification-based SAM were more consistent with the *in situ* measurements, as the points are closer to the 1:1 line in Fig. 8. For the matchups with the *in situ* measured $Z_{sd}$ of < 0.5 m on Taihu Lake, the image-estimated $Z_{sd}$ was generally higher than its corresponding field-measured values. Even for the $Z_{sd}$ values estimated from the *in situ* AHSI-band $R_{rs}$ using the proposed method, $R^2$ was only 0.42 for the 18 concurrent sampling sites on Taihu Lake, with an AURE and RMSE of 38.7% and 0.15 m, respectively. This is because when the *in situ* measured $Z_{sd}$ values were low (<0.5 m), even a small deviation, such as 0.1 m, could yield a relative error of >20%. The image-estimated $Z_{sd}$ of Taihu Lake had a lower $R^2$ (0.42) and greater AURE (58.7%) and RMSE (0.24 m), which was further caused by the errors in the image-derived $R_{rs}$. Compared to $Z_{sd}$ retrieved from the classification-based SAM using the *in situ* $R_{rs}^m$, the estimated $Z_{sd}$ of the Taihu Lake matchups using the image-derived $R_{rs}^c$ had an AURE and RMSE of 23.8% and 0.12 m, respectively. Another probable cause was larger uncertainties for the *in situ* measured $Z_{sd}$ values, which were low. In other words, $Z_{sd}$ measured by the human eye tends to have a greater relative error when that of the truth is smaller (<0.5 m).

TABLE X

ACCURACY ASSESSMENT FOR THE ESTIMATED IOPS AND $Z_{sd}$ OF THE 53 MATCHUPS BASED ON THE ZY1-02D IMAGE-DERIVED $R_{rs}^c$ BY COMPARISON WITH THE IOPS AND $Z_{sd}$ CALCULATED USING THE *In Situ* $R_{rs}^m$

| | AURE | RMSE | $R^2$ |
|---|---|---|---|
| $a$(443) | 13.2% | 0.303 | 0.829 |
| $a$(490) | 8.3% | 0.200 | 0.834 |
| $a$(532) | 8.1% | 0.127 | 0.799 |
| $a$(555) | 6.7% | 0.095 | 0.825 |
| $a$(665) | 9.4% | 0.145 | 0.760 |
| $b_b$(443) | 12.1% | 0.081 | 0.796 |
| $b_b$(490) | 11.7% | 0.076 | 0.805 |
| $b_b$(532) | 11.3% | 0.073 | 0.811 |
| $b_b$(555) | 11.1% | 0.071 | 0.814 |
| $b_b$(665) | 10.3% | 0.064 | 0.826 |
| $Z_{sd}$ | 17.9% | 0.41 | 0.98 |

The overall AURE (29.1%) for the image-derived $Z_{sd}$ based on the SAM was lower than that of the empirical model, whereas the RMSE (0.52) of the former was comparable to that of the latter (see Table IX). Specifically, for the 37 matchups in $Z_{sd}$ range of 0–3 m, the classification-based SAM produced a more accurate $Z_{sd}$ estimation than the empirical model. For the 16 matchups with $Z_{sd}$ > 3 m, the AURE and RMSE of the estimated $Z_{sd}$ using the proposed method were slightly higher than those of the empirical model.

*3) $Z_{sd}$ Estimation From ZY1-02D Images:* For pixels with an $R_{rs}(665)/R_{rs}(510)$ of ~0.55, a mixture of the $QAA_{v5}$ and $QAA_{M14}$-based SAM was applied to assure smooth transitions between the different classes of the ZY1-02D images. Specifically, we used the trial-and-error method [38] to determine the upper and lower boundaries as 0.5 and 0.56. Such boundaries ensured a smooth transition between highly turbid and moderately turbid to clear waters while exerting a small

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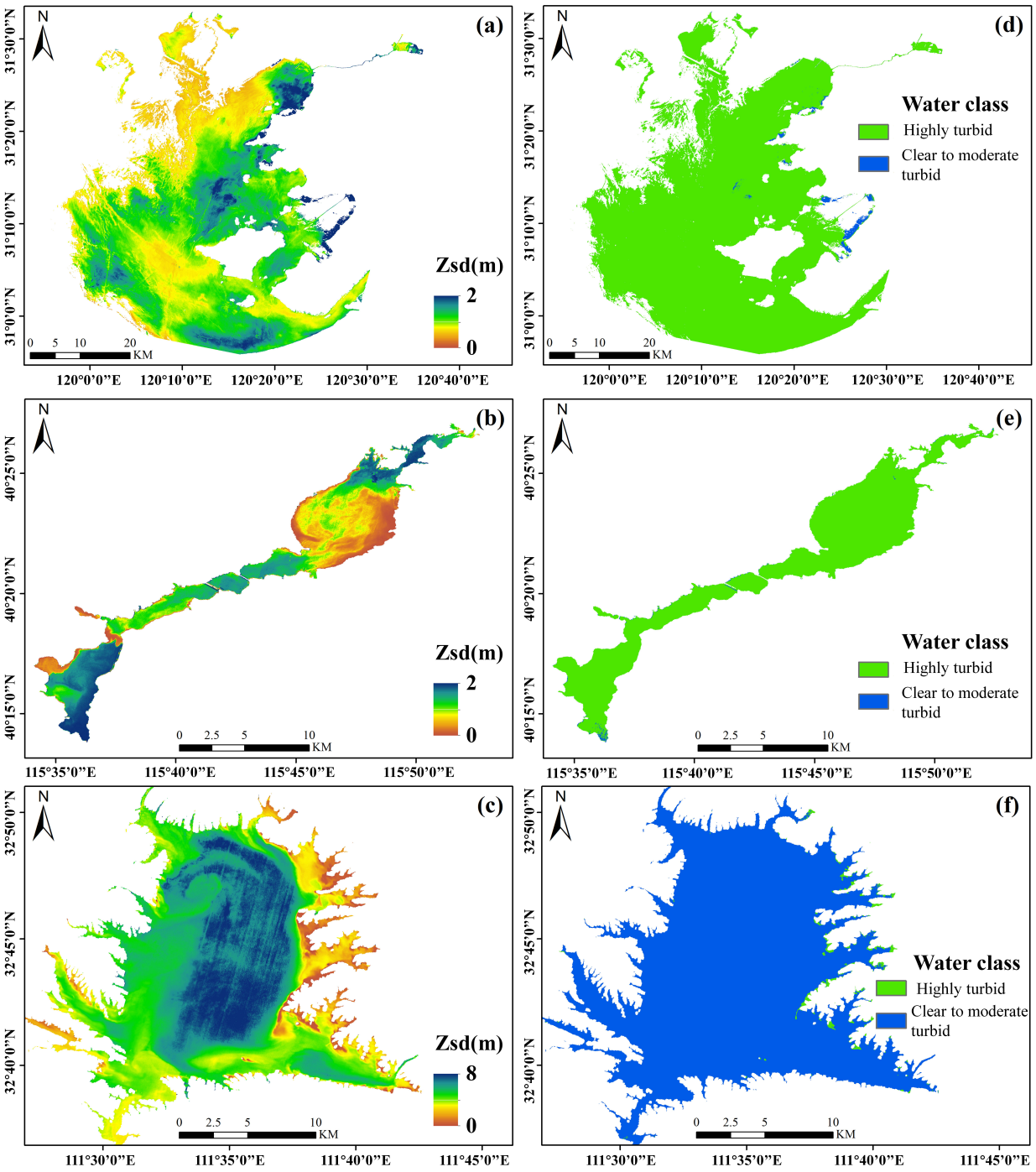



Fig. 9. Retrieved $Z_{sd}$ images from the ZY1-02D AHSI images of three study areas, including (a) Taihu Lake, (b) Guanting reservoir, and (c) Danjiangkou reservoir. Classification maps based on the hybrid QAA for (d) Taihu Lake, (e) Guanting reservoir, and (f) Danjiangkou reservoir are also provided. The $Z_{sd}$ image from the Danjiangkou reservoir is slightly affected by strip noise; however, striping did not induce large errors in the $Z_{sd}$ estimation.

impact on the $Z_{sd}$ estimations [37]. Then, for pixels with an $R_{rs}(665)/R_{rs}(510)$ between 0.5 and 0.56, $Z_{sd}$ was calculated as follows:

$$Z_{sd} = a * Z_{sd}(\text{class1}) + b * Z_{sd}(\text{class2}) \quad (9)$$

$$a = (0.56 - R_{rs}(665)/R_{rs}(510))/(0.56 - 0.5) \quad (10)$$

$$b = (R_{rs}(665)/R_{rs}(510) - 0.5)/(0.56 - 0.5). \quad (11)$$

The $Z_{sd}$ images were produced using the classification-based SAM for four concurrent ZY1-02D images (see Fig. 9). The classification maps used in the hybrid QAA are also displayed.

According to the water clarity distribution and classification maps, the estimated $Z_{sd}$ results were consistent with those in previous studies [60]–[62] on the water quality of these regions. In general, water clarity is low in Taihu Lake and the Guanting reservoir compared to the Danjiangkou reservoir. In addition, water quality in the eastern part of Taihu Lake is reportedly better than that in the west [34], which supports the relatively high $Z_{sd}$ estimated in the eastern part

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[see Fig. 9(a)]. A similar situation has been reported for the Guanting reservoir, in which higher $Z_{sd}$ values appear in southwestern areas [see Fig. 9(b)]. Some small branches of the Danjiangkou reservoir are classified as turbid [see Fig. 9(f)], which is reasonable considering that these are shallow water areas surrounded by hills.

Stripes were observed in the $Z_{sd}$ image for the Danjiangkou reservoir [see Fig. 9(c)]. Striping is a common issue for hyperspectral sensors due to their narrow bandwidth and relatively low SNR [63]. Even for sensors with broad bands (e.g., Landsat-8 OLI and Sentinel-2 MSI), this phenomenon can be notable in water scenes that have relatively weak signals [64]. Nevertheless, image noise in the Danjiangkou scene was significantly reduced using the FastHyMix algorithm [65]. This problem could be further solved by algorithm development for noise removal and radiometric correction, which is currently under investigation.

## V. Discussion

### A. Evaluation of Classification-Based SAM for $Z_{sd}$

Table VIII indicates that the image-derived $R_{rs}^{c}$ contained residual errors as compared to the *in situ* measurements. To analyze the effect of the $R_{rs}^{c}$ errors on the IOPs and $Z_{sd}$, we compared the IOPs and $Z_{sd}$ calculated using $R_{rs}^{c}$ to those retrieved based on the *in situ* ZY1-02D band $R_{rs}^{m}$. The AURE, RMSE, and $R^2$ of the IOPs were then calculated on the 53 matchups in the bands required for $Z_{sd}$ determination in (5) (i.e., the 443-, 490-, 532-, 555-, and 665-nm bands). Table X lists the accuracies for the image-derived IOPs and $Z_{sd}$.

The a and $b_b$ values estimated from the ZY1-02D image $R_{rs}$ had small AURE and RMSE values and a relatively high $R^2$, indicating that the image-derived a and $b_b$ showed general consistency with their counterparts retrieved from the *in situ* measured spectra (see Table X). The $a(443)$ and $b_b(443)$ values estimated from the image $R_{rs}$ had the highest AURE (i.e., $>12\%$). This could be due to the less accurate image-derived $R_{rs}^{c}$ in the 443-nm band (see Table VIII).

Nevertheless, the image-derived $Z_{sd}$ was consistent with $Z_{sd}$ calculated using the *in situ* $R_{rs}^{m}$, as it achieved a small AURE (17.9%) and RMSE (0.41 m) with a high $R^2$ (0.98) (see Table X). Furthermore, as revealed by Table IX, uncertainties in $R_{rs}^{c}$ introduced only minor errors in the image-estimated $Z_{sd}$ values using the classification-based SAM. This also suggests that the errors in $a(443)$ and $b_b(443)$ did not yield substantial errors in $Z_{sd}$. In contrast, for the empirical model used for comparison, its performance was linked to the accuracy of the $R_{rs}(555)$ and $R_{rs}(665)/R_{rs}(510)$ ratio, providing less satisfactory estimates compared with the classification-based SAM. Therefore, the classification-based SAM is a robust approach, which is less sensitive to $R_{rs}$ uncertainties. Moreover, unlike the empirical models, the classification-based SAM requires no measured $Z_{sd}$ for parameter tuning. It can be directly applied to both turbid and clear waters, such that it is better suited for large-scale water clarity monitoring.

The effect of the $R_{rs}^{c}$ residual error on the water type classification was also analyzed. All of the pixels in the 53 matchups were categorized in the same class determined based on the *in situ* $R_{rs}^{m}$, indicating that the water classification based on the band ratio was robust when using the ZY1-02D data.

### B. Limitations of $Z_{sd}$ Estimation Based on ZY1-02D Data

The long revisit time ($\sim$55 days) is an issue hindering the application of the ZY1-02D data for routine monitoring. However, as follows-on missions (e.g., GF5-02 and ZY1-02E) with AHSIs have been successfully launched or are planned for launch, this issue can be resolved with a hyperspectral satellite constellation.

## VI. Conclusion

The AHSI imagery from the ZY1-02D satellite can be a useful data source for water quality monitoring of inland waters. In this study, we proposed a classification-based, SAM for retrieving water clarity from the AHSI instrument onboard the ZY1-02D satellite, which is the newest generation Chinese hyperspectral mission. In this method, the water pixels were classified into clear to moderately turbid and highly turbid categories according to a red/blue ratio [i.e., $R_{rs}(665)/R_{rs}(510)$] of the ZY1-02D band $R_{rs}$. The $QAA_{v5}$ and $QAA_{m14}$ algorithms were utilized to retrieve IOPs for clear to moderately turbid water and highly turbid water, respectively. $Z_{sd}$ was then derived based on an SAM with remote sensing reflectance data and $K_d$, which was calculated from the IOPs. According to the 53 matchups with the *in situ* measurements, the image-derived $Z_{sd}$ yielded an AURE and RMSE of 29.1% and 0.52 m, respectively. Moreover, an $R^2$ of 0.98 was achieved between the image-derived and field-measured $Z_{sd}$. The retrieved $Z_{sd}$ images from the ZY1-02D data exhibited consistent distributions with the ground surveys. Our method outperformed the optimized empirical model used on the ZY1-02D data and exhibited robustness for $Z_{sd}$ retrieval, especially when considering the uncertainties in image-derived remote sensing reflectance. Therefore, the proposed method is suitable for application to ZY1-02D images for monitoring clarity in both turbid and clear waters. In addition, considering the bands applied in this classification-based SAM, it can be applied to any sensor with bands centered at 443, 490, 532, 555, 620, 665, and 708 nm.

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